Present Study reports

Report 1: Pilot Study on 10 tumor subjects, data shared by the collaborating hospital:

Purpose: The study aims to evaluate the outputs of Voxelbox, namely, language network and corticospinal tract.

Scope: This report presented a preliminary assessment of the functional integrity of eloquent networks (language) and tracts (corticospinal tract) in a clinical setting, utilizing patients with brain lesions/tumors as the study population.

Acceptance Criteria:

1. Voxelbox resting-state fMRI-based language network should be able to explain the preoperatively observed language deficits in patients with lesions - quantitative and quantitative evaluation
2. Voxelbox diffusion MRI-derived corticospinal tract should pass through the expected anatomical areas - quantitative and quantitative evaluation

Conclusions:

- The language network generated by Voxelbox has acceptable accuracy with respect to pre-operative clinical observations. Some parallels to the findings from predicate device Quicktome were also demonstrated. This along with expert clinical feedback validates Voxelbox outputs.
- The results of CST tractography, notwithstanding acceptable accuracy. The fibers accuracy and probability of occurrence was analysed in details. Discontinos fiber and extra fiber removal was performed and expert feedback was taken.

Report 2: fMRI Template coregistration report

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Purpose: The purpose is to assess the templates and template coregistration algorithm used to process resting-state fMRI in Voxelbox.

Scope: The report will explain the choice of templates used ultimately to produce different resting state networks. Further, the coregistration algorithm used for registering templates to the individual brain (healthy, dementia, and tumour) will be assessed against other common algorithms through quantitative and qualitative analyses.

Acceptance Criteria:

1. The brain templates should be based on a large number of subjects and involve task-based fMRI, among other modalities.
2. The coregistration can be assessed qualitatively by looking at whether gyral and sulcal boundaries are maintained. Moreover, areas with a known anatomical location can be confirmed using typical landmarks.
3. The coregistration can be assessed quantitatively by using 3 metrics Dice coefficient, mean squared error, and cross-correlation. The metrics can be compared between the various methods using statistics.

Conclusions:

- The templates (SENSAAS for language and HCP for other networks) used for representing various functional networks are adequate based on scientific evidence.
- SyN and SyNRA produce higher dice coefficients, which indicates better overlap between segmented structures due to their ability to capture complex deformations. Affine transformation was lower as they are less flexible in capturing intricate deformations and thus dice coefficients may be lower for detailed structures.
- Lower MSE for SyN and SyNRA due to their focus on minimizing deformations, indicating better performance.
- Higher cross-correlation for SyN and SyNRA which are deformable algorithms designed for accurate alignment and segmentation, indicating better performance.

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- SyNRA is the right choice for the implementation of the coregistration (of structural MRI and network templates) algorithm in Voxelbox
- With the implementation of SyNRA, the various network templates appear to coregister well with the healthy as well as the lesioned brain.

Report 3: fMRI processing report

Purpose: To validate the methods used for processing resting state fMRI (rs-MRI) in Voxelbox

Scope: The study will provide evidence for adequate processing methods in Voxelbox, shown in Figure 1-

- Based on previous scientific evidence
- Adequate performance based on quantitative acceptance criteria (signal-to-noise ratio)
- Adequate performance based on visual inspection

Acceptance Criteria:

The SNR post-processing should be high. The methods used should be fairly validated by previous researchers or the toolbox from where it is taken be used. There is a demonstrable effect on the data for at least a subset of the methods.

Observations:

Table 1: SNR summary statistics of the three groups

Open Neuro healthy controls dataset summary statistics

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Group	Pre SNR_openneuro	Post SNR_openneuro
Mean	9.3700	40.9890
SD	1.3109	6.1399
SEM	0.4145	1.9416
N	10	10

Tumor dataset summary statistics

Group	Pre SNR_tumor	Post SNR_tumor
Mean	16.0380	43.6780
SD	0.9777	7.0382
SEM	0.3092	2.2257
N	10	10

Dementia dataset summary statistics

Group	Pre SNR	Post SNR
Mean	17.2350	31.5840
SD	0.8036	5.0988
SEM	0.2541	1.6124
N	10	10

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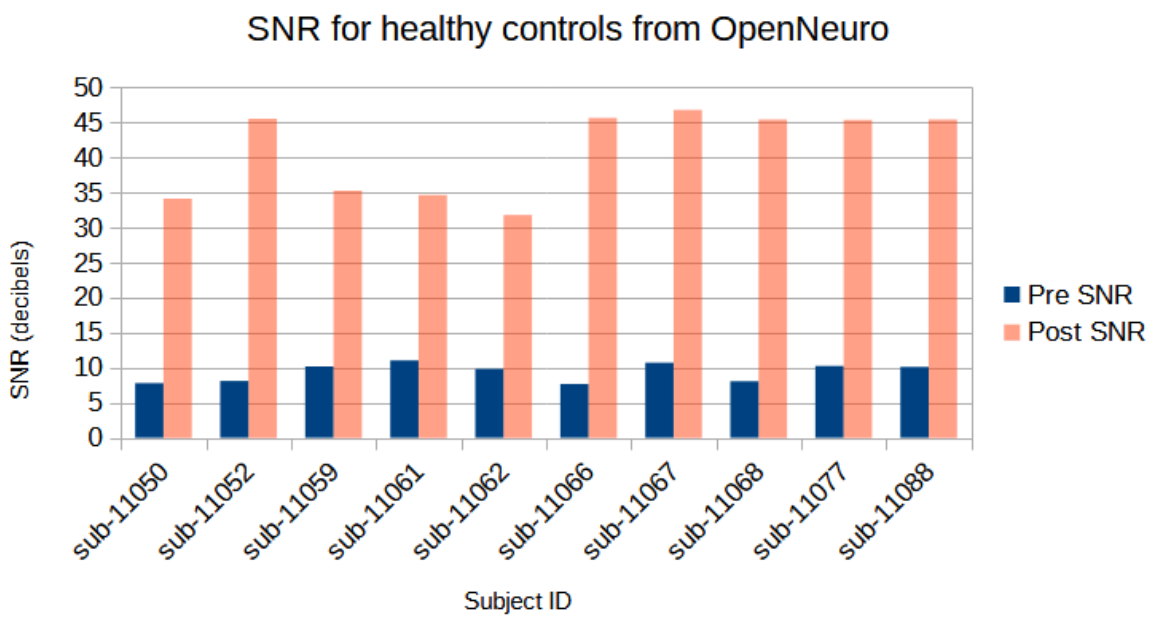

Figure 2: SNR for healthy controls from Open Neuro. The post-SNR was higher than pre-SNR (p-value < 0.0001).

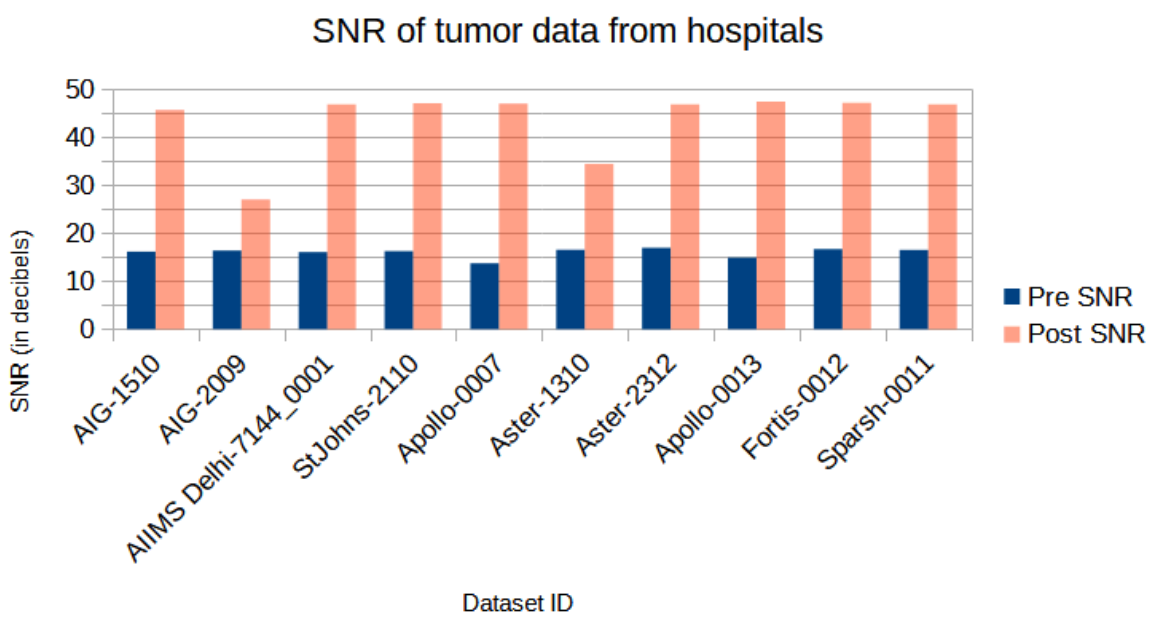

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Figure 3: SNR for tumour patients from various hospitals. The dataset ID is comprised of the hospital name followed by the Voxelbox ID. The post-SNR was higher than pre-SNR (p-value < 0.0001).

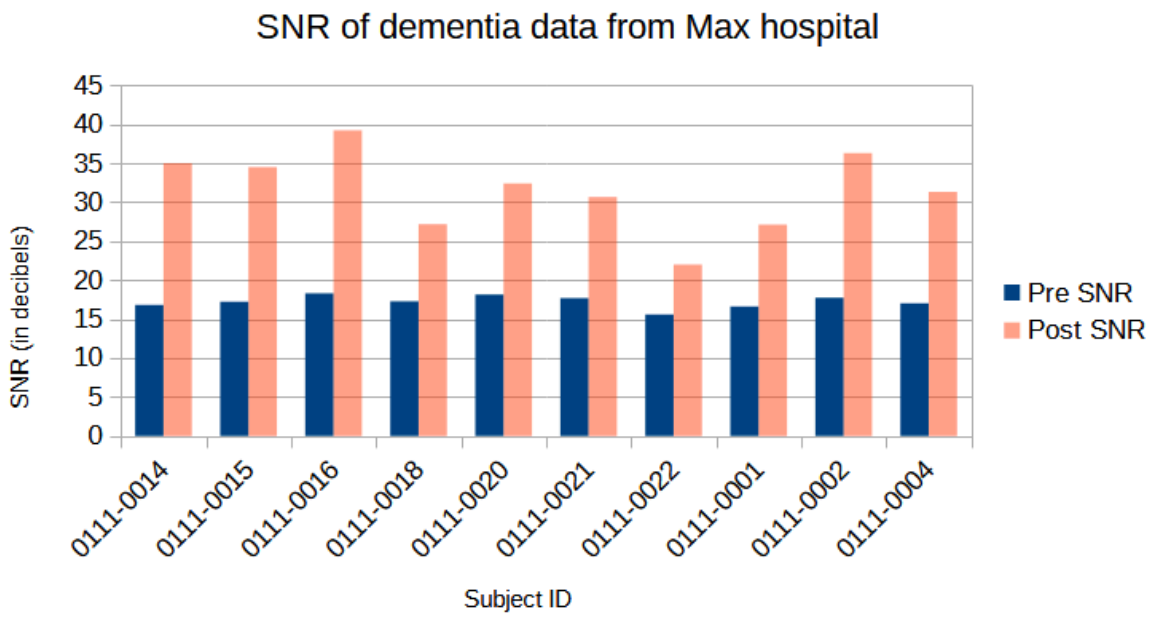

Figure 4: SNR for dementia patients from Max hospital. The post-SNR was higher than pre-SNR (p-value < 0.0001).

Conclusions:

The chosen processing methods showed good scientific evidence. The images at the intermediate processing steps demonstrated adequate processing on qualitative inspection. The SNR post-processing was significantly higher, demonstrating robust processing of resting state fMRI.

Report 4: rs-fMRI vs tb-fMRI report

Purpose: To compare language and motor networks derived from resting state fMRI (facilitated by Voxelbox) to those derived using task-based fMRI for healthy subjects.

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Scope: To compare the activations and lateralisation derived using resting-state fMRI to that generated from task-based fMRI

Acceptance Criteria:

- The clusters of rs-fMRI networks generated by VB should overlap with tb-fMRI in crucial language and motor task regions
- The lateralisation indicated by rs-fMRI and tb-fMRI should be the same for most cases

Observations:

Table 1: rs-fMRI and tb-fMRI comparison metrics (average over subjects)

Network	Sensitivity	Specificity	Distance between COMs
Language	54.1 %	98.9 %	12 mm
Motor	76.8 %	98.2 %	16 mm

Conclusions:

- On a group level, the sensitivity and specificity values for both language and motor networks were in an acceptable range given that a fixed threshold was used for tb-fMRI. The product of rs-fMRI and tb-fMRI maps indicate good topographic concordance for language. A very high concordance is not possible because tb-fMRI shows activations specific to the task, and may therefore vary according to paradigm while rs-fMRI shows activation in the entire language network. Note that this can change in tumor patients who may not have adequate task performance. Our results are similar to what has been found before. Examples, Beheshtian et al. 2021 reported: "*The concordance between rs-fMRI and sc-fMRI for the presence of dorsal somatomotor (dSMN) and ventral somatomotor (vSMN) was 75% (75 of 100 patients) and 53% (53 of 100 patients), respectively"* [3]. Abadi et al. 2017 reported- *". At the subject level, better matches between resting-state fMRI*

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and task-based fMRI were seen with an increasing order of components (53% of cases for 20 components versus 73% for 50 components)" [4].

- The distance between rs-fMRI and tb-fMRI COMs is similar to previous studies on tumor (healthy studies on rs-fMRI vs tb-fMRI are scarce) [5]. The previous study also showed that among all rs-fMRI methods, ICA-based tended to have higher COMs. **This can be improved upon in later versions of Voxelbox, however, the study used different thresholds for each subject while we used a fixed threshold for tb-fMRI.**

- The tb-fMRI produced more lateralized activations than rs-fMRI. This is similar to previous studies on tumor [6], [7]. This is not something unexpected and what makes rs-fMRI more unique. A previous study by Doucet et al. captures this beautifully (RSN= resting state network, VG= verb generation task-based)-

  "Our findings suggest that our expressive language-related RSN captures a wide set of bilateral regions which did not all appear necessary for implementing VG. Instead, the RSN may reflect a set of intrinsic, language-ready regions, likely supported by anatomical/structural connectivity present between homotopic regions "[8]

  Future versions of Voxelbox can provide affordances to tackle the inherent tendency of rs-fMRI to produce bilateral activity and lateralisation. We will be exploring different methods for laterality index (connectivity based, dynamic LI based etc) that increase concordance with tb-fMRI but also capture the uniqueness of rs-fMRI.

Report 5: Language lateralization report

Purpose: To assess various methods of deriving language lateralization using resting state fMRI

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Scope: The report assesses the calculation of the laterality index through activation and connectivity-based approaches, comparing the resulting lateralization with another non-invasive standard method, task-based fMRI.

Acceptance Criteria: The concordance of rs-fMRI and tb-fMRI derived lateralization should be high (higher performance metrics) for the method chosen to derive the language laterality index in Voxelbox

Observations:

Table 3: The performance metrics of the five rs-fMRI-based lateralisation methods compared to tb-fMRI

rs-fMRI Method	Accuracy	Precision	Recall	F1-score
New method 4- dynamic LI	0.83	0.83	0.83	0.80

Conclusions: New method 4- dynamic laterality index can be implemented in future versions of Voxelbox to derive the lateralisation of language network

Report 6: diffusion MRI processing report

Purpose: To validate the methods used for processing diffusion MRI (dMRI) in Voxelbox

Scope: The goal of this study is to demonstrate that the dMRI methods used in Voxelbox are:

- Based on previous scientific evidence
- Demonstrate adequate performance based on acceptance criteria

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Acceptance Criteria: The method should yield satisfactory results demonstrated qualitatively and/or quantitatively as defined below-

7.1. Denoising

The denoised image should be free of random alterations in intensity, also evident in image residuals, reflecting the noise removed

7.2. Gibbs artifact correction

The denoised image should be free from Gibbs ringing artifacts with the image residuals reflecting the noise removed.

7.3. B0 susceptibility distortion correction

The denoised image should clearly demonstrate the absence of peripheral distortion induced by the B0 susceptibility artifact. The mutual information of the corrected image with T1 (reference image, undistorted) should be greater than the distorted image.

7.4. Orientation correction, Motion correction and eddy current correction

It is usually difficult to see subtle motion artefacts visually in fMRI slices. By quantitatively assessing the convergence of motion and eddy current parameters through the algorithm's 5th (ideal) and 6th (extra) iterations, it is possible to deduce its effectiveness (no difference between ideal and extra iterations). Further, the 6 motion parameters (3 translational and 3 rotational) of the processed image should be closer to 0 post-processing.

7.5. Bias correction

The denoised image should be free of gradient induced by bias, more prominently evident in image residuals, reflecting the noise removed.

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7.6. Skull stripping

The skull stripping algorithm should be able to remove most parts of the skull while retaining most parts of the brain tissue

7.7. Response function and fiber orientation distribution (fOD) estimation using Constrained spherical deconvolution (CSD)

For fOD the following checks can be performed-

- White matter fODs should indeed lie within the white matter
- White matter fiber orientation distributions (fODs) align as anticipated, with red indicating the corpus callosum traversing left to right, green for the superior longitudinal fasciculus (SLF) extending posterior-anteriorly, and blue representing the corona radiata, oriented in an inferior-superior direction.

7.8. Tensor estimation (DTI)

The diffusion tensors can be best seen in FA-coloured maps where the direction of eigenvectors contributes to the colour of the map while the scalar metric fractional anisotropy (FA) value contributes to the intensity of the map. Note generally that projection fibers are blue, association fibers are green, and commissural fibers are red in a FA colored map.

7.9. Atlas-assisted ROI generation

- The atlas should be generated from a higher number of participants
- The atlas should be fairly popular in the field. There should be a fair number of papers citing it relative to the year of publication, reflecting that the atlas was deemed to be appropriate for tractography by people in the field.
- The regions of interest in the atlas should be adequate (on visual inspection) as a starting point for the generation of streamlines.

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7.10. Global Diffusion metrics generation

The images of diffusion metrics for the entire brain should look similar to the following reference images:-

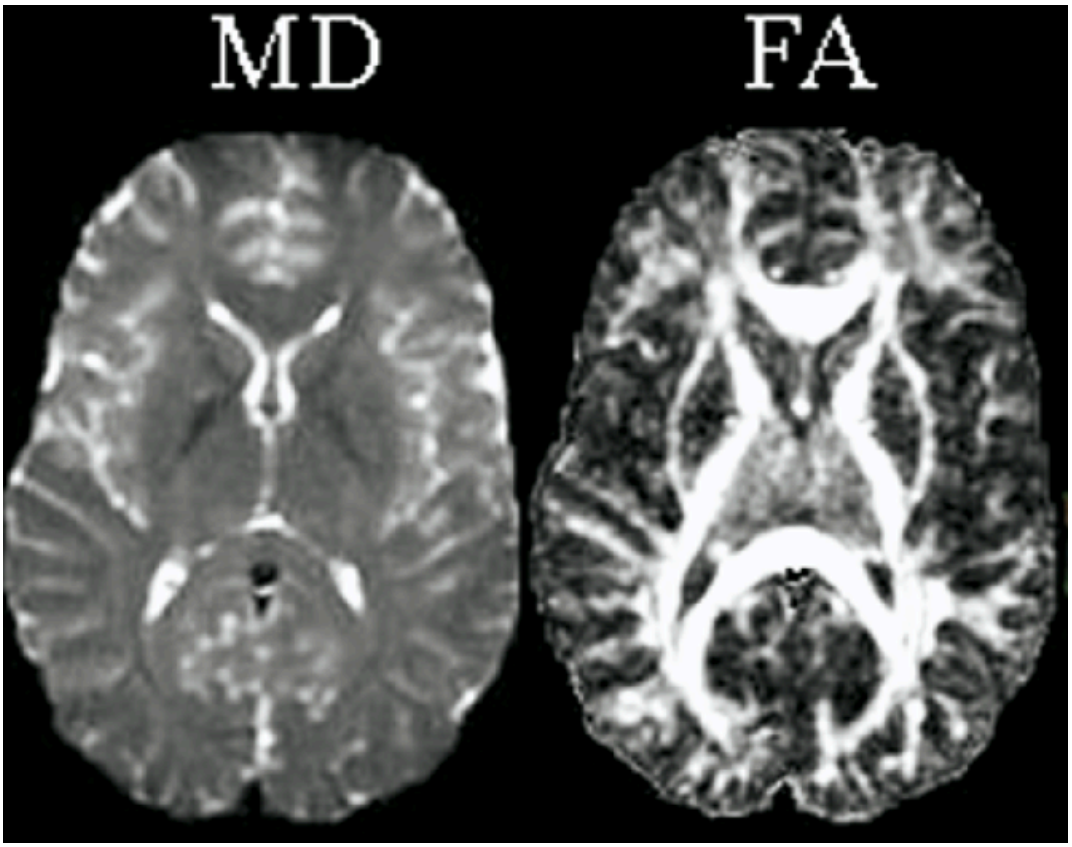

Figure 5: Representative image for scalar metric mean diffusivity (MD) and fractional anisotropy (FA). For MD, CSF appears hyperintense while in FA, white matter is hyperintense. The image has been retrieved from Alexander et al. 2007. [2]. FA maps show high intensity in the white matter, indicating high tissue coherence, and low intensity in gray matter and CSF, indicating more

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isotropic diffusion

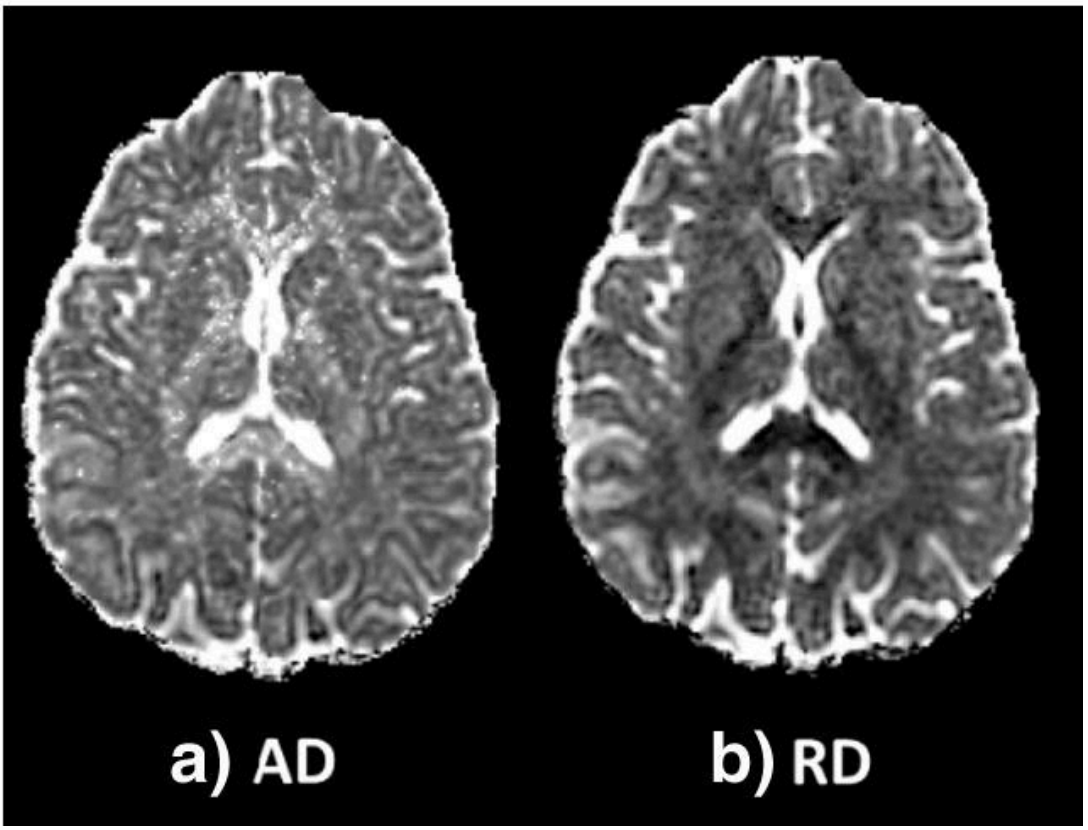

Figure 6: Representative image for scalar metric axial diffusivity (MD) and radial diffusivity (RD). AD and RD have high white matter contrast only in compact fiber tracts with known single fiber bundles, such as the corpus callosum, and the internal capsule. The image has been retrieved from Kodiweera et al. 2015. [3].

Other metrics, including sphericity, planarity, and linearity, derive from the same eigenvectors and eigenvalues. Therefore, if mean diffusivity (MD), fractional anisotropy (FA), axial diffusivity (AD), and radial diffusivity (RD) appear accurate, these metrics are likely to be reliable as well.

7.11. Fiber-specific scalar values

Some of the following checks can be performed to ensure that the scalar values have been estimated correctly.

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- Corpus callosum usually has the highest FA for a healthy person [4]. This should also be true for AD and RD which have high white matter contrast in compact fiber tracts with known single fiber bundles such as Corpus callosum.
- For a healthy brain, we expect similar values for the left and right tract.
- The value of diffusion scalar metrics can depend on a variety of factors- Un-harmonised data from different scanners can cause up to 20% variation | Different dMRI pipelines can cause variation | Surprisingly variation is found across different libraries. Nevertheless, we will check it against normative data from another source to assess the extent of deviation. Ideally, it should fall in the normative range.

Conclusions:

All 6 methods used for dMRI processing yield satisfactory results based on the defined acceptance criteria. For the 4 algorithms where only a qualitative evaluation was used, another layer of validation can be added through quantitative metrics. Further, a larger sample size can be used for each metric to demonstrate consistent performance, especially in edge cases.

Report 7: Probabilistic tractography report

Purpose: The purpose of this study is to assess the tractography algorithm used in Voxelbox through a literature review as well as a review of the tracts generated using the algorithm.

Scope:

- Comparison of probabilistic tractography with its popular alternative deterministic tractography
- Review of probabilistic tractography validation, against DES or histopathological validation in human subjects
- Review the scope or various applications of probabilistic tractography
- Description of probabilistic tractography algorithm with Voxelbox parameters
- Semi-quantitative review of tracts generated for a healthy brain

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- Qualitative review of tracts generated for a tumour brain

  Acceptance Criteria:

  - The algorithm chosen for tractography viz., probabilistic shows advantages over its popular alternative deterministic tractography based on the literature review.
  - The algorithm chosen for tractography viz., probabilistic has been validated against the gold standard of electrical stimulation based on literature review.
  - The algorithm chosen for tractography viz., probabilistic has demonstrated utility for a variety of diseases or disorders
  - The tracts generated using Voxelbox are acceptable based on feedback from clinician experts and internal inspection according to what is known and accepted in the field

Conclusions:

- DTI-based deterministic fiber tracking was found to have limited sensitivity, particularly for the lateral aspects of the corticospinal tract.

- CSD (in combination with Probabilistic tractography) was found to be more robust for mapping hyperdirect pathways, important for deep brain stimulation.

- For complex fiber configuration using phantom, probabilistic tractography was shown to have a high false positive rate and stringent connectome thresholding was recommended if probabilistic tractography was to be employed for connectome mapping.

- Probabilistic tractography was better at reconstructing Meyer's loop than deterministic tractography

- Probabilistic tractography reconstructed the arcuate fasciculus more completely and performs better through areas of tumour and/or edema.

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- Probabilistic tractography showcased use in a wide variety of diseases including tumour, epilepsy, traumatic brain injury, Alzheimer's disease, essential tremors, akinesia, and amyotrophic lateral sclerosis.

- Probabilistic tractography showed better spread and fanning fibers near the motor cortex

- CSD showed a higher rate of true positives and a slight increase in false positives, with no significant differences in true or false negatives. Of the probabilistic algorithms, CSD results were superior to others. The most sensitive detection of fiber bundles belonging to the CST was achieved when using the CSD method was used in combination with a probabilistic algorithm, whereas the deterministic DTI method, representing the standard method used in current neurosurgical navigational systems, was less sensitive and produced false-negative results

- Probabilistic methods outperformed deterministic, qualitatively shown by better spread of fibers, tract lengths closer to anatomy, and quantitatively by better sensitivity and specificity. More accurate models for fiber estimation, such as QBall and CSD when combined with probabilistic methods provide an improved solution to the above-mentioned problems. The use of probabilistic DTI algorithms has demonstrated improved sensitivity and predictive power in determining the course of white matter fibers.

- An additional advantage of using CSD is the ability to perform a detailed whole-brain tractography allowing for the visualization of fibers without the need for locally predefined seed and target regions, thus revealing the whole extent of white matter complexity and its interaction with pathological structures.

- **Voxelbox uses probabilistic tractography with CSD modelling, effectively the best combination of modelling and fiber tracking methods. The literature appraisal showed that the choice of methods for Voxelbox tractography is appropriate.**

- **Based on the Internal review- The fibers were grossly anatomically correct, but not precise in some cases. Some general problems throughout all cases were**

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missing fibers, and extra fibers, especially evident with corpus callosum fibers being a part of any fiber bundle that spread across both hemispheres. There was a discontinuity in the fibers which could potentially be attributed to denoising algorithms. The possible mitigation strategies to improve tractography are the introduction of constrained tractography like anatomically constrained tractography (ACT), pruning, improvement of diffusion models, and improvements on the denoising algorithm.

- **Based on the External Review-** Most of the questions on whether the tracts ended at functional activation were marked True by the clinician. In some cases, only part of the motor or language tract ended in their respective activation/deemed anatomical cortical location. Therefore, the tracts were correlated to the functional activations. Most of the responses on the correctness of the structural + functional system were also 1 or correct.

- **False activations and discontinuity (fragmentation), identified via internal review were also evident in some of the nuanced feedback given by clinicians on morphology as well as when comparing to GE.** The missing fibers/imprecise fibers, found during the internal inspection were not very evident in the external feedback. It was only found in a few cases.

- The neurosurgeon seemed to be more satisfied with the tract performance compared to the neuroradiologist. In many cases across tracts where VB showed better performance compared to GE. **The Voxelbox tractography was found to be more sensitive, robust, continuous and specific in a subset of cases.**

- The table below summarizes the tract-wise performance on part 1 and part 3 of the external feedback. As part 2 was positive overall and internal review appears to be a subset of external feedback, part 1 and part 3 of the external feedback will govern the overall performance assessment. Based on this, it appears that **the performance of Voxelbox tractography for CST, OR, FAT, IFOF, and ILF is good with some scope for improvement, especially with respect to false/extra fibers.**

- **The performance of AF, SLF, and CC are satisfactory with a lot of scope for improvement.** For AF and CC, the issue is with both extra fibers and missing fibers,

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both due to denoising algorithm. SLF is a large tract composed of three different components and generating all of them together has been difficult. Most of these will be taken up in future versions of Voxelbox.

Overall Voxlebox tractography outputs are satisfactory as per the acceptance criteria.

Table 5: Summary of part 1 and part 3 of external feedback

Tracts of interest	Average performance on morphological correctness	Summary of GE comparison
CST	93 %	100 % Equivalent or better compared to GE | more robust, continuous, sensitive, and specific
AF	80 %	60 % Equivalent or better in 60 % | False or extra fibers
OR	70 %	80 % Equivalent or better in 80 % | False or extra fibers
FAT	100 %	Better than GE in 2 cases where feedback was provided
IFOF	90 %	90 % Equivalent or better than GE
SLF	100 %	Inconsistent performance compared to GE, has been marked better, suboptimal, and equivalent in equal number of cases
ILF	83 %	77 % Equivalent or better than GE
CC	75 %	Equivalent or suboptimal compared to GE

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Report 8: Revamped Diffusion MRI and tractography pipeline

Purpose: The study validates the incremental changes that were implemented in the dMRI pipeline in several versions from v2.0.3 to v3.2.0.

release Version ID	pipeline Version ID	Date of release	changes made to dmri pipeline	Implementation
v2.0.3	v4.6.9_internal	5-Jul-2023	DMRI pipeline made modular; DMRI QA QC Images updated;	Robust skull stripping for dMRI. Results in dicom format which are neuronavigation compatible Providing scalar values such as Avg. FA, ADC, MD, AD, CL, CS, CP for both the hemispheres in results Output overlays on T1w/T1c/T2w/FLAIR present in results.
	v4.6.9_internal	26-July-2023	Extract normative data class is added	
		26 Oct2023	deepbet to skull stripping	
		11 Jan2023	unified packaging started	
		28-Jan-2024	deletion of intermediate files	
v2.1.0		11-01-2024	Robust skull stripping for dMRI	
	v4.8.4	8-Feb-2024		
v3.1.0	v5.0.1	27-March-2024	merged new ACT pipeline	
		28-March-2024	added denoise MPPCA	
	v5.0.0	8-Apr-2024		
	v5.0.1.2	18/04/2024	modification of version for prodcution	
v3.2.0	v5.2.0	18-Apr-2024	Major Changes	dMRI: Extra Fibre Removal. dMRI: Robust Denoising. dMRI: Diffusion Matrix updated. dMRI: Discontinuous fiber fixed. Flat Folder Structure for results. Optimized storage space consumption 2 Atlas HCP and Xtract provided for CST Tract Only Eloquent Network and Associated tracts in output, reports and viewer for a given hospital

Scope: The report assesses the changes made to the dMRI pipeline in the new version of Voxelbox in pre-processing (Denoising and Skull stripping) as well as processing, notably- Generation of FOD and response function estimation (1), probabilistic atlas instead of average (2), functional anisotropy-based refined seeding method (3), length constraints (4), and anatomical constraints (5) on probabilistic tractography, ensemble tractography (6), and topologically informed pruning (7). The report also shows how the new pipeline is a mitigation to the risk posed by the presence of extra fibers in the tractography output (risk ID R030).

Acceptance Criteria: Most processes will be validated first through scientific validation. The novel changes should be backed by good rationale, logic, and a discussion of potential risks or limitations. Important processes should show adequate changes expected through the algorithm either in the quantitative metrics (for example, SNR in denoising, length distribution with or without implementation of length constraints) and/or visually in outputs (for example, Generation of FOD, the effect of functional anisotropy-based refined seeding method, the effect of pruning).

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- Overall, the new tracts should show increased accuracy as well as specificity (reduction in extra fibres) compared to the old pipeline.
- The new tracts should also be validated by clinicians and not increase false negatives at the expense of increased accuracy.
- The changes in tract properties such as length and volume (if any) should be in line with changes in the pipeline

Conclusions:

Internal review

- Compared to the old algorithm, the outputs from the new algorithm do not contain extra fibers from CC that could previously be seen in fiber bundles like CST, IFOF, etc.
- SLF has been split into SLF-I, SLF-II, and SLF-III for easy viewing and better rendering and understanding of the tract.
- The base of CST is thicker for both HCP and Xtract template-based tractography. This can be addressed in future versions.
- There are still some extra fibers and noise in a few cases likely because as tracts in general are slightly thicker, the noise is coming as a byproduct. This can be addressed in future versions. It is important to note that extra fibers can also occur due to suboptimal (noisy) raw clinical data.
- In cases with bigger tumors, the tracts seem to be thinner on the side of the tumor, likely because the probability of seeding is less on that side. This will be taken care of in the next version
- The tracts in the old pipeline were discontinuous, likely due to the denoising algorithm. The new tracts seem to be continuous

External review: **This part will be updated after the St John's study is complete**

Summary of evaluation of changes in the dMRI pipeline

Method: Denoising	Satisfactory as per specific acceptance criteria
Method: Skull stripping	Satisfactory as per specific acceptance criteria

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Method: Multi-shell response function estimation and FOD generation with skull-stripping mask	Satisfactory as per rationale and specific acceptance criteria
Method: Generation of seed regions	Satisfactory as per rationale and global acceptance criteria (internal and external review)
Method: Extraction of normative data	Satisfactory as per rationale, global acceptance criteria (internal and external review), and changes in tract length.
Method: Anatomical constraints	Satisfactory as per literature, rationale, global acceptance criteria (internal and external review), and changes in tract volume
Method: Ensemble tracking	Satisfactory as per literature, rationale, global acceptance criteria (internal and external review), and changes in tract volume
Method: Track filtering	Satisfactory as per literature, rationale and global acceptance criteria (internal and external review)
Metric: Tract properties	Satisfactory as per acceptance criteria

Overall, the new tracts should show increased accuracy as well as specificity (reduction in extra fibres) compared to the old pipeline. The new tracts were validated by clinicians. The changes in tract properties were in line with changes in the pipeline.

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Report 9: dMRI Template coregistration report

Purpose: The purpose of this study is to assess and validate the coregistration of white matter template and structural MRI image used in the dMRI pipeline

Scope: The report evaluates the white matter template coregistration step of the dMRI pipeline in tumor and TBI groups, using healthy controls as a reference, through analytical and clinical validation

Acceptance Criteria:

- **Qualitative clinical**: The coregistration between white matter template and T1 in healthy, tumour and TBI cases should be grossly acceptable based on visual inspection.
- **Quantitative clinical**: The dice coefficient between coregistered white matter template to T1 and coregistered white matter tract to T1 in tumour and TBI cases should not exceed those achieved for healthy cases by a large margin.
- The end output- tracts should be accurate upon visual inspection despite poor coregistration.

Conclusions:

On visual inspection, it appears that the template masks are not accurately fitted after coregistration in all cases. However, in most cases, this does not affect tractography. This is likely because the masks serve only as seeding regions, determining where tractography begins. It is the fiber orientation distribution that appears to influence the tract length and how far the tracts extend. Therefore, the 1st and 3rd acceptance criteria are fulfilled.

Upon quantitatively assessing coregistration, the Dice scores (which indicate the fit of coregistration) for tumor and TBI cohorts appear to be generally within the same range as those of healthy controls. While tumor and TBI subjects tend to have more occurrences of lower Dice scores, some tracts in healthy subjects also exhibit poor coregistration scores.

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Interestingly, despite brain deformities in patients, even in cases with low Dice scores, the tracts were generally well reconstructed.This suggests that tractography is somewhat independent of errors introduced by template-to-subject brain coregistration. Therefore, the 2nd and 3rd acceptance criteria are fulfilled. It is also worth noting that subsequent steps in the tractography pipeline such as anatomical constraints introduced through shell-masking (refer to this dMRI report, Change-6, for details), can also introduce errors, particularly due to an additional coregistration step required for shell alignment. Nevertheless, the overall benefits of automatic tractography i.e., eliminating manual effort, significantly reducing processing time, and enabling the generation of multiple tracts outweighs these errors (if any).

To improve tractography further and to avoid errors induced by coregistration , template-agnostic tractography methods will be developed to eliminate the need for coregistration. For example, whole-brain tractography can be generated and later segmented into individual tracts. This approach is commonly used in state-of-the-art toolboxes like DSI Studio, FreeSurfer, and 3D Slicer. However, these methods are typically unsupervised or require substantial amounts of training data, making their development and optimization resource-intensive. Another potential mitigation strategy is to use a machine learning-based approach for coregistration, where the atlas is directly identified from the white matter segmentation of the subject's brain. This method would also require training data, which could include automatically generated template coregistration masks for healthy brains (which tend to be more accurate) and potentially manually coregistered and/or annotated tumor brains. An example of such an implementation is [8].

To conclude, based on the above findings, template coregistration generally performs acceptably, both in absolute terms and in its impact on subsequent tractography.

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Report 10: dMRI scalar metrics report

Purpose: This study aims to validate the scalar maps and corresponding tract-specific scalar metrics, produced as an output of Voxelbox against their intended use i.e., insights for rich reporting in various neurological disorders.

Scope: The study evaluates the diffusion MRI (dMRI) scalar metrics produced by VoxelBox based on predefined acceptance criteria from clinical and scientific evidence from the literature. The fractional anisotropy values are chosen for inspection but the results should be generalizable across all scalar metrics such as axial diffusivity, radial diffusivity, and mean diffusivity as they are derived from the same base values. The scalar metrics are derived from scalar maps produced by Voxelbox by averaging the scalar values across the tract streamlines. For details of processing and derivation, the reader is directed to a previous report.

Acceptance Criteria: FA on an individual level for clinical insights: The scalar metrics should reflect WM integrity at an individual level to feed into insights for rich reporting. FA on a group level for research insights: The findings within and between groups should be explainable from a clinical or scientific perspective and should align with current research and field knowledge. Overall, these criteria are indicative of the potential of scalar metrics in rich reporting by clinicians, especially neuroradiologists.

Conclusions:

FA on an individual level for clinical insights

For each tract, a few tumour subjects show asymmetry in FA values that is evident on visual inspection with the healthy/contralesional side having higher FA than the ipsilesional side. This is in line with our hypothesis of reduced FA on the tumour side [2]. Different tracts may be affected in different subjects owing to different tumour locations. Many subjects show similar FA values on the left and right side, potentially indicating intact fiber integrity on the left hemisphere, where the tumour was present. On the other hand, all healthy subjects show a symmetrical pattern. This is in line with our hypothesis.

In dementia, fewer subjects show differential FA for the two hemispheres and the direction of asymmetry differs for different subjects, unlike tumours. This is contrary to our assumption of FA symmetry in dementia. This may indicate the clinician of differential

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neurodegeneration patterns with the side having less FA potentially indicative of lesser damage.

Overall FA values are informative at an individual level, however, to increase the sensitivity of FA, FA can be shown in smaller regions of interest within the tract and as a function of distance with respect to the lesion/tumour/neurodegeneration site. The Voxelbox viewer facilitates a closer inspection of FA for precise clinical insights.

FA on a group level for research insights

No significant differences were found intra-hemispherically for the Dementia and Healthy groups. This was expected and in line with our hypothesis. For the tumour group, FA was significantly lower for the ipsilesional side compared to contralesional, for AF but not for other tracts. These findings show that FA changes were specific to tumour effects rather than normal variation.

It is possible that AF was consistently affected across all cases, however, since the tumour lobar location and tracts disrupted owing to the tumour are not known, this is inconclusive. For example, CST is likely to be impacted if the tumour location is in sensorimotor regions. The tumour in frontal regions could impact tracts like FAT and IFOF and the tumour in temporal regions may impact AF and SLF.

Significant differences between healthy and tumour patient groups were found: healthy individuals were found to have higher FA values across five tracts (CC, AF, OR, IFOF, SLF) compared to ipsilesional sides in tumour patients. This has been reported in the literature before [3]. According to the literature, AF is extremely prone to damage caused by diseases such as stroke and brain tumour. Researchers have also pointed out that DTI metrics such as FA should be used. For brain tumours, it can be precisely positioned before surgery to significantly improve the quality of life of patients after surgery, and secondly, it can be used to predict neurological function recovery, such as language and cognition after surgery [5]. As expected, the tumour didn't affect the contralesional side with FA values closer to those of healthy subjects.

There also exists clear individual variability in tumour response, the degree of FA value reduction doesn't seem to be uniform across subjects, even for tumours in similar locations.

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This suggests individual factors (genetics, overall brain health) might influence how WM tracts respond to tumour presence.

AF connects Broca's AF area with Wernicke's area and plays an important role in verbal function, and reduced FA values have been reported in the literature, especially in frontotemporal dementia [6]. This finding is in concordance with our results. In our study, we also observed significant differences in FA between healthy and dementia for CC. According to the introduction section of a paper on dementia, several studies did not observe any changes in FA, while at the same time, other studies have shown reduced FA in AD subjects in posterior regions of the CC or anterior region of the CC [7]. The same study found lower FA values in AD compared to healthy controls in the genu, body, and splenium of the CC and similar but smaller clusters for the AD– Mild Cognitive Impairment (MCI, early stage of AD) comparison. This may mean that WM microstructural changes in the CC occur later in the pathological cascade. Similar to the study, our study considered datasets from different types of dementia from multiple sources.

Most of the results captured expected patterns in line with our hypotheses/assumptions based on the literature, dMRI processing methods utilized in VoxelBox for producing scalar metrics seem valid and effective. Even though only tumour and dementia datasets were considered, we expect Voxelbox outputs to be useful across other neurological disorders as well as for testing hypotheses related to white matter integrity. For example, FA reduction was also evident in patients with Parkinson's disease when compared to the healthy cohort [8]. Similarly, in Temporal Lobe Epilepsy, FA was found to be reduced compared to healthy control groups [9]. In conclusion, scalar metrics (like FA) have the potential to provide valuable insights into neurological disorders, thus helping clinicians such as neuroradiologists in rich reporting.

Future Study report(s)

Report 11: Validation of rsfMRI based image derived phenotypes

Purpose: The purpose of this study is to explain and validate resting state fMRI results other than networks generated using Voxelbox, i.e., image-derived phenotypes or IDPs.

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Scope: The report covers the following results derived from resting state fMRI-

- The amplitude of low-frequency fluctuation (ALFF) and the fractional amplitude of low-frequency fluctuation (fALFF)
- Regional homogeneity (ReHO)
- Functional connectivity (FC)
- Graph theory-based metrics- degree, global efficiency, modularity, clustering coefficient, small worldness, hub of the network

Acceptance Criteria: The methods should demonstrate adequate performance based on scientific evidence and qualitative inspection.

- These metrics are useful in research, as demonstrated through the literature review.
- ReHO- Certain regions like DMN will show higher ReHO in baseline subjects.
- ALFF, fALFF: (I) normative plots, with lit reference, see which regions show higher ALFF. (II) Power spectrum of processed rs-fMRI time series before and after derivation of ALFF, fALFF.
- For FC, average all FC metrics from HC, find out from literature how normative FC looks like, correlation and anti-correlation between areas, compare with known facts, and show areas with higher FC.
- Comparison of our outputs to DPABI.
- Showcase a successful research use (in alignment with our intended use): Statistically compare the IDPs between schizophrenia and healthy controls and see if our findings correspond to any of the meta-analyses we reviewed, Python, and other toolboxes.

Conferences and Publications

Conference abstracts

Conference	Title and Link
Bernstein Conference 2020	Diagnosis of Mild-Alzheimer's disease using multiple brain parcellation ensemble machine learning with

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	resting-state fMRI
FENS 2020	VoxelBox Plus - A Novel Tool For Dynamic Network Localization Of Brain Functional Activity In Dementia Using Resting State - fMRI: Implications On Detection, Treatment And Management
Nature conference 2022	DEEP LEARNING-BASED HEAD MOTION ESTIMATION FOR CORRECTING BETWEEN-VOLUME MOTION ARTIFACTS IN DIFFUSION MR IMAGES
RSNA 2022	A novel connectomic analysis framework for personalized presurgical planning using structural MRI, rs-fMRI, and DWI Part II
RSNA 2022	Multimodal brain tumour segmentation using deep convolutional neural networks
ISNR 2023	Exploring the utility of artificial intelligence augmented functional connectomics approach in brain tumours- a prospective cohort study
NSICON 2023	Utilising resting state fMRI and dynamic functional connectivity for language lateralisation Presentation
RSNA 2023 (accepted, chose not to attend)	Does resting-state fMRI have the potential for presurgical functional mapping?
RSNA 2023 (accepted,	NODDI vs traditional DTI metrics RSNA

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chose not to attend)	2023_Educational exhibit
ECR 2023 (accepted and attended)	Harmonizing tb-fMRI and rs-fMRI: A Generative approach for mapping Language Networks
RSNA 2024 (accepted and attended)	Effective Glioblastoma segmentation through tractography
RSNA 2024 (accepted attended)	Exploring the utility of an artificial intelligence-augmented functional connectomics approach in brain tumours to evaluate the involvement of non-eloquent areas—a prospective cohort study
RSNA 2024 (accepted attended)	Accelerated Motion Correction with Deep Learning for Functional MRI - A potential for real-time fMRI processing for faster clinical assessment
RSNA 2024 (accepted attended)	Advancing Pediatric Brain Mapping: An AI-Driven Adaptive Functional MRI Pipeline for Mapping of Functional Networks
OHMB 2025 (accepted and plan to attend)	2908 - Adaptive AI Approach to Pediatric fMRI Network Mapping
OHBM 2025 (accepted and plan to attend)	2796 - Mapping Tumor Interfaces with Brain Networks: A Tractography-Based Surgical Planning Tool
OHBM 2025 (accepted and plan to attend)	2867 - A Python-Based Toolkit for Advanced White Matter Tractometry and Voxel-Level Connectivity

Papers

Towards artificial intelligence in mental health by improving schizophrenia prediction with multiple brain parcellation ensemble-learning

https://www.nature.com/articles/s41537-018-0070-8

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Extending schizophrenia diagnostic model to predict schizotypy in first-degree relatives

https://www.nature.com/articles/s41537-020-00119-y

Translational Connectomics: overview of machine learning in macroscale Connectomics for clinical insights

https://bmcneurol.biomedcentral.com/articles/10.1186/s12883-024-03864-0

Regulations

1. ISO 13485--Quality Management System Certification - Approved
2. ~~Test License from CDSCO (Indian Regulatory Body) - Aproved~~
3. SOC2 compliance report - Available
4. Manufacturing License – Approved
5. HIPAA certification - Approved
6. ISO 27001 ISMS -- Ongoing
7. FDA -- Ongoing

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